Arnab Maity

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PROFILE

Final year Computer Science undergraduate specializing in Full Stack Development and Generative AI engineering. Experienced in building scalable web applications, RESTful APIs, and implementing high-performance graph algorithms. Hands-on freelance and internship experience delivering production-ready frontend and full-stack solutions.

EDUCATION

Degree	Specialization	Institute	Year	CGPA/%
B.Tech	Computer Science & Engineering	Sister Nivedita University	2022–2026	7.6/10
HS	Science	Gopalnagar Biharilal Vidyapith	2022	91%
Secondary	–	Gopalnagar Biharilal Vidyapith	2020	70%

TECHNICAL SKILLS

- **Generative AI & Core:** Generative AI Integration, Graph Theory, High-Performance Computing (OpenMP), Data Structures & Algorithms (C++)
- **Backend Architecture:** Node.js, Express.js, REST APIs, JWT Authentication, Role-Based Authorization
- **Frontend Engineering:** React.js, Next.js, JavaScript, HTML5, CSS3, Redux Toolkit, Tailwind CSS
- **Databases & Storage:** MongoDB, PostgreSQL, Database Indexing
- **Cloud & DevOps:** Docker, AWS (EC2, S3), Git, GitHub, Linux

EXPERIENCE & FREELANCE

Web Development Intern [Oasis Infobyte (AICTE OIB-SIP)] **May 2025 – Jun 2025**

- **UI Development:** Engineered reusable React.js components, improving rendering performance by 25% across 4 web applications.
- **Frontend Optimization:** Streamlined asset delivery and DOM structures, reducing initial page load times by over 300ms.
- **Version Control:** Collaborated using Git workflows, managing branch merges and conflict resolutions across multi-developer repositories.

Freelance Frontend Engineer [Client Projects] **Jan 2025 – Present**

- **High-Fidelity Implementation:** Designed and deployed responsive React interfaces for client requests, ensuring 100% cross-browser compatibility.
- **Performance SEO:** Optimized client web applications for SEO and Core Web Vitals, driving a 20% increase in mobile user engagement.

KEY PROJECTS

Hospital Management System **(Full Stack / MERN Stack)**

- **Tech Stack:** React.js, Node.js, Express.js, MongoDB, Redux Toolkit, JWT, Tailwind CSS, Nodemailer.
- **Role-Based Access Control:** Architected a full-stack platform supporting RBAC across 4 distinct user tiers (Admin, Doctor, Patient, Receptionist).
- **Automated Scheduling Engine:** Built an appointment booking system with real-time slot validation, preventing scheduling conflicts across 100+ daily requests.
- **Security & State Management:** Integrated JWT authentication and bcrypt hashing for secure REST APIs, utilizing Redux Toolkit to reduce state sync latency by 35%.
- **Automated Workflows:** Configured Nodemailer integrations for automated transactional emails (booking confirmations, cancellations, password resets).

RESEARCH PROJECTS

Pebble Traversal-Based Fault Detection for Digital Microfluidic Biochips **(Algorithms & C++)**

- **Tech Stack:** C++, OpenMP, Graph Theory, Algorithms.
- **Defect Detection Pipeline:** Formulated graph-based pebble traversal algorithms in C++ to automate defect identification on DMFB arrays.
- **Dynamic Rerouting:** Applied Manhattan distance heuristics to dynamic rerouting algorithms, bypassing faulty biochip nodes with zero execution deadlocks.
- **Parallel Execution:** Parallelized path generation algorithms using OpenMP multi-threading, accelerating fault traversal computation by 40% over sequential execution.